Heeta Bhanushali

heetabhanushali@gmail.com heetabhanushali.vercel.app heetabhanushali

+91 8591156784

ABOUT ME

I'm a Master's student in Software Systems Engineering at RWTH Aachenwith a Computer Engineering background from DJ Sanghvi. My interests have moved toward the business side of technology, especially startups and building companies. I'm looking for a place to learn by doing.

WORK EXPERIENCE

Transline Air Cargo Services Pvt. Ltd., *Software Engineer Intern* 06/2024 – 11/2024 | Mumbai, India

- Designed relational database schema replacing Excel-based operations across shipment tracking, custom documentation, invoicing, and client management
- Built ETL pipelines to migrate and clean historical data from unstructured spreadsheets
- Enforced business rules through database-level constraints and developed automated reporting

EDUCATION AND TRAINING

RWTH Aachen, *Masters in Software Systems Engineering* 10//2026 – Present | Aachen, Germany

Dwarkadas J. Sanghvi College of Engineering 10/2022 – 06//2026 | Mumbai, India
Bachelors in Technology in Computer Engineering

SKILLS

Languages (C, C++, Java, Python, Rust, Dart, JavaScript) | **Frameworks & Tools** (FastAPI, Flask, Flutter, Firebase, Supabase, GitHub Actions, Docker, SQLAlchemy) | **Concepts** (Compiler Design, Data Structures & Algorithms, REST APIs, CI/CD, WebAssembly, Encryption, Microservices, Static Analysis, Finite State Machines, Code Optimization)

LANGUAGE SKILLS

Kutchhi (Mother tongue(s)) | **Gujarati** (Mother tongue(s)) | **Marathi** | **Hindi** | **English** | **German**

PROJECTS

TransitCore — Formally-Grounded Workflow Engine 12/2025 – 02/2026

- Built a workflow engine enforcing declarative state transitions with a custom expression language and EBNF grammar
- Implemented static verifier proving temporal logic properties with atomic transitions and automatic rollback
- Developed REST API with optimistic locking, audit trails, auto-generated diagrams, and 93% test coverage

Technologies: Python, Lark (EBNF parser generator), SQLAlchemy, Graphviz, Hypothesis

MiniLang Compiler & Static Analyzer 02/2025 – 05/2025

- Built a complete compiler with lexer, parser, type checker, optimizer, and C code generator

- Implemented four optimization passes (constant folding, dead code elimination, propagation, strength reduction)
- Designed static analysis engine with complexity metrics and deployed WebAssembly browser playground with Monaco Editor

Technologies: Rust, WebAssembly, wasm-bindgen, Logos, miette, serde, Monaco Editor, Vercel

Gloves Shop — E-Commerce Microservices Platform 12/2024 – 01/2025

- Built 8 polyglot microservices across 12 containerized services with database persistence and async messaging
- Implemented CI/CD pipelines with automated builds, image publishing, and infrastructure-as-code for cloud deployment
- Configured real-time monitoring and alerting dashboards with container-level metrics

Technologies: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Prometheus, Grafana, Node.js, Python, Java, Go, PHP, MongoDB, MySQL, Redis, RabbitMQ

Cross-Platform Packet Sniffer & Network Security Monitor 09/2024 – 11/2024

- Built a real-time network analyzer with packet inspection across 13 protocols (Layers 2–7) and heuristic-based threat detection
- Designed multi-threaded capture engine with concurrent parsing, statistics aggregation, and cross-platform support
- Developed interactive dashboard with live packet streaming, protocol charts, and security alerts

Technologies: Python, Scapy, Flask, Multithreading, REST APIs, JavaScript, Chart.js, HTML/CSS

HelloChat – Real-time Social Chat Application 12/2023 – 05/2024

- Built an offline-first encrypted chat app with end-to-end encryption and cross-device key recovery
- Engineered offline message queue with exponential backoff, deduplication, and bidirectional sync
- Developed real-time messaging, typing indicators, read receipts, soft-delete conversations, and AI chatbot integration

Technologies: Flutter, Dart, Supabase, PostgreSQL, Hive, AES-256-GCM, WebSocket

PUBLICATIONS

Bias Mitigation in Language Models using a Multi-Dimensional Evaluation and Comparison 01/12/2024

ICRMLAI'25

Authors: Arham Shah, Jai Jain, Heeta Bhanushali, Granth Jain, Nilesh Patil, Chinmay Raut

TADR: Typology-Aware Dynamic Routing for Zero-Shot Cross-Lingual Transfer 01/12/2024

CSCT 2025

Authors: Heeta Bhanushali, Jai Jain, Ameya Suryawanshi, Meera Narvekar, Kranti Ghag, Khushali Deulkar